Factorized τ posterior

one 1-D grid per process precision instead of a 3-D grid

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Setting: τ learned on a grid

Process error, per pool:

$$Q = \mathrm{diag}\left(\frac{1}{\tau_1},\ \frac{1}{\tau_2},\ \frac{1}{\tau_3}\right)$$

Each grid point $\tau^{(g)}$ runs its own mixture filter; its weight is updated by

$$\pi_t(g) \ \propto\ \pi_{t-1}(g)\ p\left(y_t \mid \tau^{(g)}, y_{1:t-1}\right)$$

3-D product grid: $G_1 G_2 G_3$ filters $(16^3 = 4096)$

Factorized: one block per axis

Block a: τ_a on its own grid, the others at plug-in values

$$\tau^{(a,g)} = \big(\tau_a^{(g)},\ \hat{\tau}_{-a}\big), \qquad \hat{\tau}_b = \frac{1}{\mathbb{E}_{\pi_b}[1/\tau_b]} = \Big(\sum_g \frac{\pi_b(g)}{\tau_b^{(g)}}\Big)^{-1}$$

$$\pi_{a,t}(g) \ \propto\ \pi_{a,t-1}(g)\ p\Big(y_t \mid \tau^{(a,g)}, y_{1:t-1}\Big)$$

$G_1 + G_2 + G_3$ filters $\qquad (16 \cdot 3 = 48)$

Factorized: what it assumes, and how it summarizes

Assumes the posterior is close to a product, and each conditional is near its plug-in:

$$p(\tau \mid y_{1:t}) \;\approx\; \prod_{a=1}^{3} p(\tau_a \mid y_{1:t}), \qquad p(\tau_a \mid y_{1:t}) \approx \pi_{a,t}$$

State and scenario summaries average the three blocks:

$$p(x_t, k \mid y_{1:t}) \;\approx\; \frac{1}{3} \sum_{a=1}^{3} \sum_{g} \pi_{a,t}(g)\; p\Big(x_t, k \mid \tau^{(a,g)}, y_{1:t}\Big)$$

Metrics: is the joint posterior a product?

From the 3-D grid posterior $\pi_t(i,j)$ of each pair (τ_a, τ_b):

$$\rho_{ab} = \frac{\mathrm{Cov}_\pi(\log \tau_a, \log \tau_b)}{\sqrt{\mathrm{Var}_\pi(\log \tau_a)\ \mathrm{Var}_\pi(\log \tau_b)}}$$

$$I(\tau_a; \tau_b) = \sum_{i,j} \pi(i,j)\ \log \frac{\pi(i,j)}{\pi_a(i)\,\pi_b(j)} = \mathrm{KL}\big(\pi_{ab} \,\|\, \pi_a \pi_b\big)\ \geq 0$$

$$\mathrm{TV}\big(\pi_{ab},\ \pi_a \pi_b\big)\ \leq\ \sqrt{I/2} \qquad \text{(Pinsker)}$$

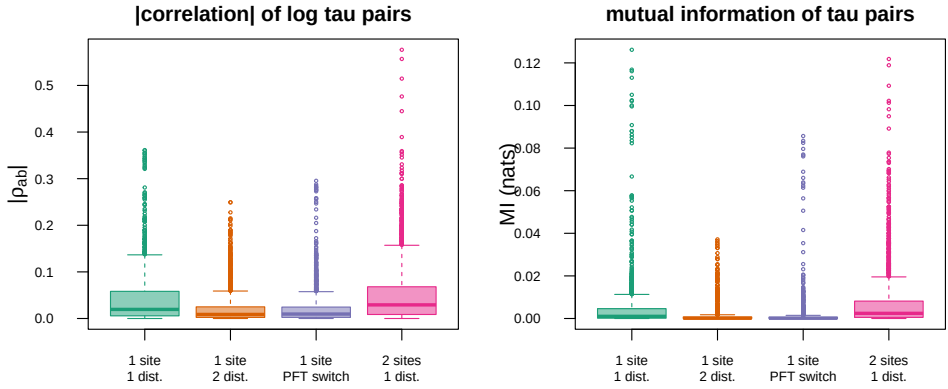
$\log \tau$: the grid and prior are log-uniform; ρ is scale-free. $I = 0$ iff independent (nats).

Result 1: the τ pairs are nearly independent

21 seeds $\times$ 25 years $\times$ 3 pairs per experiment, 3-D grid posterior

	$\lvert\rho_{ab}\rvert$		$I(\tau_a;\tau_b)$ (nats)	
experiment	median	max	median	max
1 site, 1 dist.	0.02	0.36	0.001	0.126
1 site, 2 dist.	0.01	0.25	0.000	0.037
1 site, PFT switch	0.01	0.30	0.000	0.086
2 sites, 1 dist.	0.03	0.58	0.002	0.122

Result 1: every seed, year and pair



Median $|\rho| \leq 0.03$, median $I \leq 0.003$ nats. Largest $I = 0.13$ nats $\Rightarrow$

$\mathrm{TV}(\pi_{ab}, \pi_a \pi_b) \leq \sqrt{0.13/2} = 0.25$.

Metrics: does the factorized filter agree with the 3-D grid?

Same set-up and importance samples for both filters:

$$\Delta_P = \max_t \left| P_{\text{fact}}(k \neq 0 \mid y_{1:t}) - P_{\text{3-D}}(k \neq 0 \mid y_{1:t}) \right|$$

$$\Delta_{\sigma,a} = \left| \log \frac{\sigma_a^{\text{fact}}}{\sigma_a^{\text{3-D}}} \right|, \qquad \sigma_a = \sqrt{\mathbb{E}[1/\tau_a \mid y_{1:T}]}$$

$$z_a = \frac{\Delta_{\sigma,a}}{\text{sd}_{\text{3-D}}(\log \sigma_a \mid y_{1:T})}$$

Baseline: the same Δs between two 3-D grid runs with different importance samples (the filter's own Monte Carlo noise). $\Delta_\sigma = 0.1$: a 10% difference; z: in posterior sds.

Result 2: 9–16× faster

experiment	grid points factorized	grid points 3-D	speed-up
1 site, 1 dist.	48	4096	14×
1 site, 2 dist.	36	1728	11×
1 site, PFT switch	36	1728	9×
2 sites, 1 dist.	36	1728	16×

Speed-up: median ratio of filter run times (3-D / factorized), 21 seeds.

Result 3: disturbance probabilities agree to within Monte Carlo noise

experiment	Δ_P: fact. vs 3-D median	Δ_P: fact. vs 3-D max	baseline: 3-D vs 3-D median	baseline: 3-D vs 3-D max
1 site, 1 dist.	0.003	0.129	0.004	0.218
1 site, 2 dist.	0.001	0.024	0.002	0.022
1 site, PFT switch	0.002	0.055	0.003	0.025
2 sites, 1 dist.	0.002	0.080	0.002	0.101

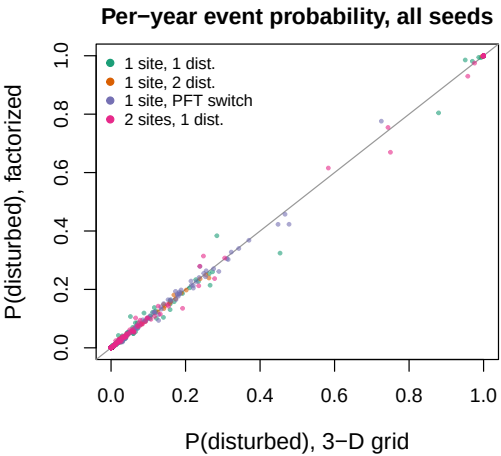
21 seeds per experiment; max over the 25 years of each run.

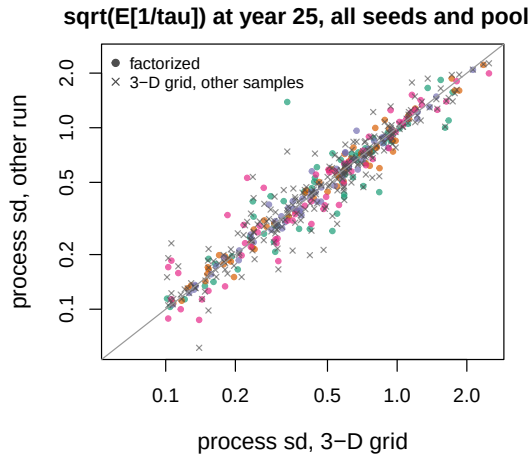
Result 4: process sd differs no more than Monte Carlo noise

experiment	Δ_σ: fact. vs 3-D median	max	baseline: 3-D vs 3-D median	max	z median	> 1
1 site, 1 dist.	0.12	1.42	0.13	0.81	0.18	3%
1 site, 2 dist.	0.07	0.34	0.06	0.36	0.08	0%
1 site, PFT switch	0.04	0.37	0.04	0.30	0.05	0%
2 sites, 1 dist.	0.15	0.86	0.16	0.82	0.21	0%

21 seeds $\times$ 3 pools per experiment, at year 25.

Results 3–4: every seed





Left: P(disturbed) per year. Right: σ_a at year 25, per pool and seed. Line: equality.

Numerical health

Factorized filter, 21 seeds $\times$ 4 experiments $\times$ 25 years (`experiments/synthetic/dropped-across-seeds`):

- forecast mass dropped (collapsed weights or singular covariance): **1** step of 2100, mass 5×10^{-248} (PFT switching, seed 5)
- grid points left with zero mass after dropping: **none**

Summary

- The joint τ posterior is close to a product: median $|\rho_{ab}| \leq 0.03$, $I \leq 0.13$ nats.
- Factorized vs 3-D grid differences (Δ_P, Δ_σ) are the size of the differences between two 3-D grid runs with other importance samples.
- Process-sd differences are small against the posterior width: median $z \leq 0.21$; $z > 1$ in 2 of 252 seed–pool cases.
- Cost: $\sum_a G_a$ instead of $\prod_a G_a$ grid points, 9–16$\times$ faster.